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import pandas as pd
import math

# Load dataset
df = pd.read_excel("id3_dataset.xlsx")
df.head()

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	Feature_1	Feature_2	Feature_3	Feature_4	Feature_5	Feature_6
0	High	Medium	High	High	Very High	High
1	Very High	Medium	Very High	High	Medium	Low
2	Low	Medium	Low	High	Low	Low
3	High	High	High	High	Low	Medium
4	High	Very High	Low	Medium	High	Very High

	Feature_8	Feature_9	Feature_10	Feature_11	Feature_12	Target
0	Low	Medium	Very High	Very High	High	Class_B
1	High	Medium	High	Medium	High	Class_B
2	Low	Very High	Very High	Very High	Low	Class_B
3	High	Medium	Very High	Medium	High	Class_B
4	Very High	Very High	Medium	Medium	Low	Class_B

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def entropy(column):
    values = column.value_counts(normalize=True)
    return -sum(values * values.apply(lambda x: math.log2(x)))

def information_gain(df, feature, target):
    total_entropy = entropy(df[target])

    values = df[feature].unique()

    weighted_entropy = 0
    for v in values:
        subset = df[df[feature] == v]
        weighted_entropy += (len(subset)/len(df)) *
entropy(subset[target])

    return total_entropy - weighted_entropy

def id3(df, features, target):
    # 1. If pure class → return label
    if len(df[target].unique()) == 1:
        return df[target].iloc[0]

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# 2. If no features left → majority class
if len(features) == 0:
    return df[target].mode()[0]

# 3. Compute gain for all features
gains = {f: information_gain(df, f, target) for f in features}

# 4. Choose best feature
best_feature = max(gains, key=gains.get)

tree = {best_feature: {}}

# 5. Split dataset
for value in df[best_feature].unique():

    subset = df[df[best_feature] == value]

    if subset.empty:
        tree[best_feature][value] = df[target].mode()[0]
    else:
        remaining_features = [f for f in features if f !=
best_feature]
        tree[best_feature][value] = id3(subset,
remaining_features, target)

    return tree

target = "Target"

features = list(df.columns)
features.remove(target)

decision_tree = id3(df, features, target)

print("FINAL ID3 DECISION TREE:\n")
print(decision_tree)

```

FINAL ID3 DECISION TREE:

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{'Feature_3': {'High': {'Feature_9': {'Medium': {'Feature_1': {'High':
'Class_B', 'Very High': {'Feature_6': {'Low': {'Feature_2': {'Very
High': 'Class_A', 'Low': 'Class_B'}}}, 'Medium': 'Class_B', 'Very
High': 'Class_B', 'High': 'Class_B'}}}, 'Medium': 'Class_A', 'Low':
{'Feature_12': {'Very High': 'Class_A', 'Medium': {'Feature_2':
{'Low': 'Class_A', 'Very High': 'Class_B'}}}, 'High': 'Class_B', 'Low':
'Class_B'}}}}, 'Low': {'Feature_1': {'High': {'Feature_10': {'Medium':
'Class_B', 'Very High': 'Class_A', 'Low': 'Class_A', 'High':
'Class_B'}}}, 'Very High': {'Feature_8': {'High': 'Class_A', 'Medium':
'Class_B', 'Very High': 'Class_B'}}}, 'Low': {'Feature_12': {'Very
High': 'Class_B', 'Low': 'Class_B', 'Medium': 'Class_A'}}}, 'Medium':
'Class_B'}}}, 'High': {'Feature_10': {'High': {'Feature_12': {'Low':

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'Class_B', 'Very High': 'Class_B', 'High': 'Class_A', 'Medium': 'Class_A'}}}, 'Very High': 'Class_A', 'Medium': {'Feature_7': {'Low': 'Class_B', 'Medium': 'Class_A', 'High': 'Class_B', 'Very High': 'Class_A'}}}, 'Low': 'Class_B'}}}, 'Very High': {'Feature_2': {'Very High': {'Feature_1': {'Very High': {'Feature_4': {'High': 'Class_A', 'Low': 'Class_B', 'Medium': 'Class_B'}}}, 'High': 'Class_B', 'Medium': 'Class_B', 'Low': 'Class_A'}}}, 'Medium': {'Feature_5': {'High': 'Class_A', 'Medium': 'Class_A', 'Low': 'Class_B', 'Very High': 'Class_B'}}}, 'Low': {'Feature_10': {'Very High': 'Class_A', 'Low': 'Class_B', 'Medium': {'Feature_11': {'High': 'Class_A', 'Medium': 'Class_A', 'Very High': 'Class_B'}}}, 'High': 'Class_A'}}}, 'High': 'Class_A'}}}}, 'Very High': {'Feature_10': {'High': {'Feature_11': {'Medium': 'Class_B', 'High': {'Feature_8': {'High': 'Class_A', 'Low': 'Class_B', 'Very High': {'Feature_2': {'Low': 'Class_B', 'High': 'Class_A'}}}, 'Medium': 'Class_B'}}}, 'Very High': {'Feature_1': {'High': 'Class_B', 'Very High': {'Feature_4': {'Medium': 'Class_A', 'Very High': 'Class_A', 'High': 'Class_B'}}}, 'Low': {'Feature_4': {'High': 'Class_A', 'Low': 'Class_B'}}}, 'Medium': 'Class_A'}}}, 'Low': {'Feature_1': {'High': 'Class_A', 'Very High': 'Class_B', 'Medium': 'Class_A'}}}}, 'Very High': {'Feature_11': {'Low': {'Feature_5': {'Low': {'Feature_2': {'Low': 'Class_B', 'High': 'Class_B', 'Very High': 'Class_A'}}}, 'High': 'Class_A', 'Medium': 'Class_A'}}}, 'High': 'Class_B', 'Very High': 'Class_B', 'Medium': 'Class_B'}}}, 'Medium': {'Feature_11': {'Very High': {'Feature_6': {'Medium': 'Class_A', 'Very High': 'Class_B', 'Low': 'Class_B'}}}, 'Medium': 'Class_A', 'High': {'Feature_6': {'Very High': 'Class_B', 'Medium': 'Class_A', 'High': 'Class_B', 'Low': 'Class_B'}}}, 'Low': {'Feature_1': {'Very High': 'Class_A', 'Low': 'Class_B'}}}}, 'Low': {'Feature_8': {'Medium': {'Feature_2': {'Medium': {'Feature_9': {'Low': 'Class_A', 'Medium': 'Class_A', 'Very High': 'Class_B'}}}, 'Low': 'Class_A', 'High': 'Class_B', 'Very High': 'Class_B'}}}, 'Very High': {'Feature_5': {'Medium': 'Class_B', 'High': 'Class_A', 'Low': 'Class_A'}}}, 'High': 'Class_B', 'Low': 'Class_A'}}}}, 'Low': {'Feature_9': {'Very High': {'Feature_12': {'Low': {'Feature_4': {'High': 'Class_B', 'Medium': 'Class_B', 'Low': {'Feature_1': {'Medium': 'Class_A', 'Very High': 'Class_B', 'Low': 'Class_A'}}}, 'Very High': 'Class_B'}}}, 'Medium': {'Feature_1': {'High': {'Feature_6': {'Medium': 'Class_B', 'High': 'Class_A'}}}, 'Medium': 'Class_B', 'Low': 'Class_A', 'Very High': 'Class_A'}}}, 'High': {'Feature_11': {'High': 'Class_A', 'Very High': 'Class_A', 'Low': {'Feature_5': {'High': 'Class_A', 'Medium': 'Class_B'}}}, 'Medium': 'Class_B'}}}, 'Very High': {'Feature_6': {'Low': 'Class_B', 'High': 'Class_B', 'Medium': 'Class_A', 'Very High': 'Class_A'}}}}, 'Medium': {'Feature_11': {'Very High': {'Feature_2': {'Low': 'Class_A', 'High': 'Class_B', 'Medium': 'Class_B'}}}, 'High': 'Class_B', 'Low': {'Feature_4': {'Low': 'Class_B', 'Medium': {'Feature_1': {'Very High': 'Class_B', 'Low': 'Class_A'}}}, 'High': 'Class_A', 'Very High': 'Class_B'}}}, 'Medium': {'Feature_1': {'Medium': 'Class_A', 'High': {'Feature_6': {'Low': 'Class_B', 'Very High': 'Class_A', 'Medium': 'Class_A'}}}, 'Low': 'Class_B'}}}}, 'Low':

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{'Feature_2': {'Very High': {'Feature_7': {'High': 'Class_B', 'Low': 'Class_A', 'Very High': 'Class_A', 'Medium': 'Class_A'}}, 'High': 'Class_A', 'Low': {'Feature_6': {'Low': 'Class_A', 'High': 'Class_B', 'Medium': {'Feature_4': {'High': 'Class_A', 'Very High': 'Class_B'}}, 'Very High': 'Class_B'}}, 'Medium': {'Feature_5': {'High': 'Class_A', 'Medium': 'Class_A', 'Low': {'Feature_6': {'High': 'Class_B', 'Medium': 'Class_A', 'Low': 'Class_A'}}, 'Very High': 'Class_B'}}}}, 'High': {'Feature_8': {'Medium': {'Feature_7': {'Very High': 'Class_A', 'Medium': {'Feature_6': {'Very High': 'Class_B', 'High': 'Class_A', 'Medium': 'Class_B'}}, 'High': 'Class_A'}}, 'Very High': {'Feature_6': {'Low': 'Class_A', 'High': 'Class_B'}}, 'Low': {'Feature_1': {'Medium': 'Class_B', 'Very High': 'Class_B', 'High': 'Class_A'}}, 'High': {'Feature_1': {'High': 'Class_B', 'Very High': 'Class_B', 'Medium': 'Class_A', 'Low': 'Class_B'}}}}}}, 'Medium': {'Feature_1': {'Low': {'Feature_10': {'High': 'Class_B', 'Low': {'Feature_9': {'Medium': 'Class_B', 'High': 'Class_A', 'Low': {'Feature_2': {'High': 'Class_A', 'Low': 'Class_B', 'Very High': 'Class_A', 'Medium': 'Class_B'}}, 'Very High': 'Class_A'}}, 'Very High': {'Feature_8': {'Low': 'Class_B', 'Very High': 'Class_B', 'High': 'Class_A', 'Medium': 'Class_B'}}, 'Medium': {'Feature_7': {'Low': {'Feature_6': {'Medium': 'Class_A', 'High': {'Feature_2': {'Low': 'Class_A', 'High': 'Class_A', 'Very High': 'Class_B'}}, 'Very High': 'Class_B', 'Low': 'Class_A'}}, 'High': {'Feature_4': {'Medium': 'Class_A', 'Low': 'Class_A', 'Very High': 'Class_B', 'High': 'Class_A'}}, 'Medium': {'Feature_2': {'Medium': 'Class_A', 'Very High': 'Class_B', 'Low': 'Class_B'}}, 'Very High': {'Feature_9': {'High': 'Class_B', 'Low': 'Class_A', 'Medium': 'Class_A', 'Very High': 'Class_B'}}}}, 'High': {'Feature_11': {'Very High': {'Feature_7': {'Medium': {'Feature_2': {'Medium': 'Class_A', 'Low': 'Class_B', 'Very High': 'Class_B'}}, 'High': 'Class_A', 'Very High': 'Class_B', 'Low': 'Class_A'}}, 'High': {'Feature_6': {'Medium': 'Class_A', 'Very High': 'Class_B', 'High': 'Class_A', 'Low': 'Class_A'}}, 'Low': {'Feature_6': {'Low': 'Class_A', 'Very High': 'Class_A', 'High': 'Class_B', 'Medium': 'Class_A'}}, 'Medium': {'Feature_7': {'Low': 'Class_B', 'Very High': {'Feature_2': {'Medium': 'Class_A', 'High': 'Class_B'}}, 'High': 'Class_B', 'Medium': 'Class_A'}}}}, 'Very High': {'Feature_6': {'Low': {'Feature_8': {'High': 'Class_A', 'Medium': 'Class_B', 'Very High': 'Class_B', 'Low': {'Feature_9': {'Very High': 'Class_B', 'High': 'Class_A'}}}}, 'Very High': {'Feature_5': {'Medium': 'Class_B', 'Low': 'Class_B', 'Very High': {'Feature_8': {'Medium': 'Class_A', 'Low': 'Class_A', 'Very High': 'Class_B'}}, 'High': 'Class_A'}}, 'Medium': {'Feature_5': {'Low': 'Class_A', 'Very High': 'Class_B', 'High': 'Class_A', 'Medium': 'Class_A'}}, 'High': 'Class_A'}}}}}}
```

```
def print_tree(tree, indent=""):
    if not isinstance(tree, dict):
        print(indent + "→ " + str(tree))
        return
```

```

for feature, branches in tree.items():
    print(indent + f"[ {feature} ]")
    for value, subtree in branches.items():
        print(indent + f"├─ {value}")
        print_tree(subtree, indent + " | ")

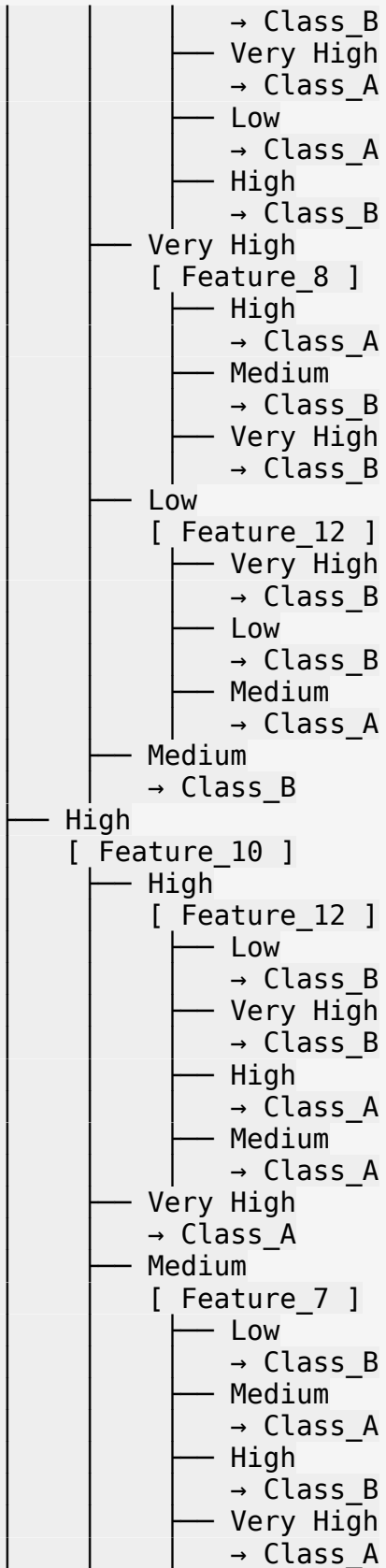
```

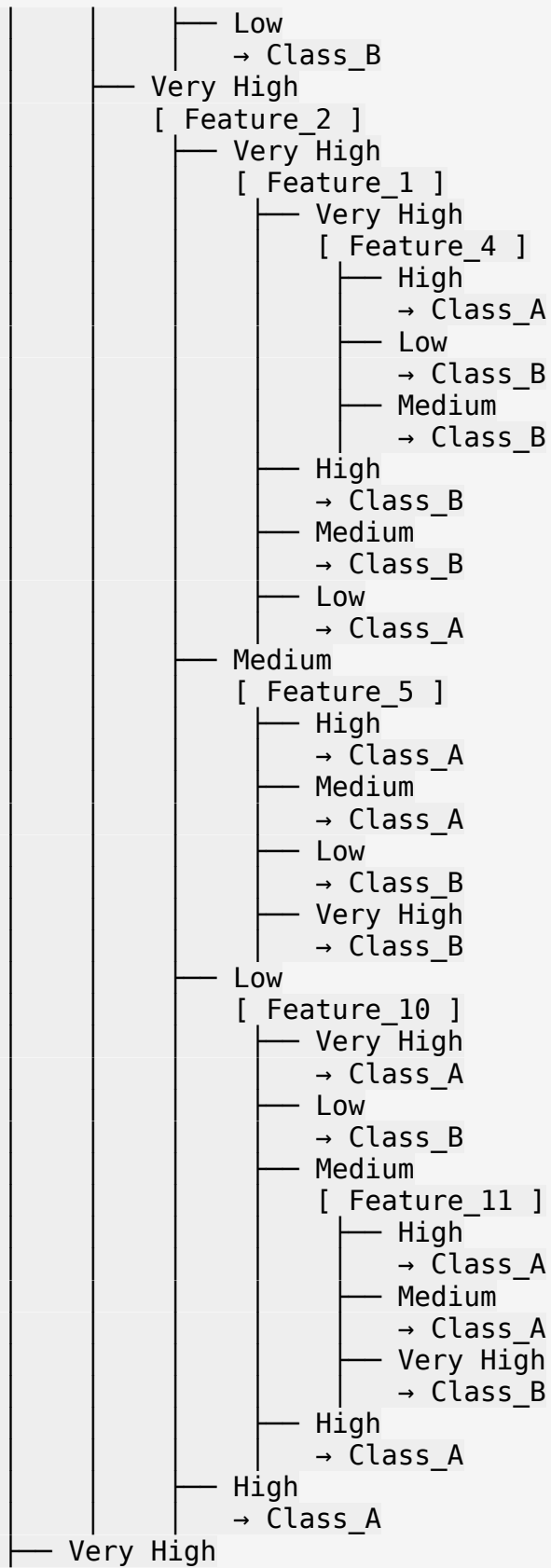
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print_tree(decision_tree)
```

```

[ Feature_3 ]
├─ High
│   └─ [ Feature_9 ]
│       └─ Medium
│           └─ [ Feature_1 ]
│               └─ High
│                   → Class_B
│               └─ Very High
│                   └─ [ Feature_6 ]
│                       └─ Low
│                           └─ [ Feature_2 ]
│                               └─ Very High
│                                   → Class_A
│                               └─ Low
│                                   → Class_B
│                       └─ Medium
│                           → Class_B
│                       └─ Very High
│                           → Class_B
│                       └─ High
│                           → Class_B
│               └─ Medium
│                   → Class_A
│               └─ Low
│                   └─ [ Feature_12 ]
│                       └─ Very High
│                           → Class_A
│                       └─ Medium
│                           └─ [ Feature_2 ]
│                               └─ Low
│                                   → Class_A
│                               └─ Very High
│                                   → Class_B
│                       └─ High
│                           → Class_B
│                       └─ Low
│                           → Class_B
│           └─ Low
│               └─ [ Feature_1 ]
│                   └─ High
│                       └─ [ Feature_10 ]
│                           └─ Medium

```





```
[ Feature_10 ]
├── High
│   ├── [ Feature_11 ]
│   │   ├── Medium
│   │   │   └── → Class_B
│   │   ├── High
│   │   │   ├── [ Feature_8 ]
│   │   │   │   ├── High
│   │   │   │   │   └── → Class_A
│   │   │   │   ├── Low
│   │   │   │   │   └── → Class_B
│   │   │   │   └── Very High
│   │   │   │       ├── [ Feature_2 ]
│   │   │   │       │   ├── Low
│   │   │   │       │   │   └── → Class_B
│   │   │   │       │   └── High
│   │   │   │       │       └── → Class_A
│   │   │   │       └── Medium
│   │   │   │           └── → Class_B
│   │   └── Very High
│   │       ├── [ Feature_1 ]
│   │       │   ├── High
│   │       │   │   └── → Class_B
│   │       │   ├── Very High
│   │       │   │   ├── [ Feature_4 ]
│   │       │   │   │   ├── Medium
│   │       │   │   │   │   └── → Class_A
│   │       │   │   │   ├── Very High
│   │       │   │   │   │   └── → Class_A
│   │       │   │   │   └── High
│   │       │   │   │       └── → Class_B
│   │       │   │   └── Low
│   │       │   │       ├── [ Feature_4 ]
│   │       │   │       │   ├── High
│   │       │   │       │   │   └── → Class_A
│   │       │   │       │   └── Low
│   │       │   │       │       └── → Class_B
│   │       │   │       └── Medium
│   │       │   │           └── → Class_A
│   │       └── Low
│   │           ├── [ Feature_1 ]
│   │           │   ├── High
│   │           │   │   └── → Class_A
│   │           │   ├── Very High
│   │           │   │   └── → Class_B
│   │           │   └── Medium
│   │           │       └── → Class_A
│   └── Very High
│       └── [ Feature_11 ]
```

